

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4.       | (a) |  | <p><b>Requirements</b></p> <ul style="list-style-type: none"> <li>• Direction of polarisation in common</li> <li>• Comparable amplitude</li> <li>• Same frequency / wavelength</li> <li>• Coherent</li> <li>• Diffraction</li> </ul> <p><b>Possible explanation material</b></p> <ul style="list-style-type: none"> <li>• Coherent sources means constant [<b>or</b> constant or zero] phase difference between sources</li> <li>• Sketch-graphs showing meaning of constant phase difference</li> <li>• Rapidly shifting interference pattern if phase difference varies</li> <li>• No cancellation if polarisation directions or amplitudes different</li> <li>• Diffraction needed for overlap</li> </ul> <p><b>Possible examples of compliance and non-compliance</b></p> <ul style="list-style-type: none"> <li>• Two <i>slits</i> illuminated by single laser would fulfil requirements</li> <li>• Slits illuminated by ordinary lamp or separate lamps wouldn't</li> <li>• Contrived cases e.g. sources polarised at 90° to each other; polaroids crossed or one source (e.g. slit) covered by dark glass acceptable</li> </ul> | 6               |     |     | 6     |       |      |